Working in $GF(2^8)$

$m(x) = x^8 + x^4 + x^3 + x + 1$

the irreducible modulus poly in $GF(2^8)$

$\Rightarrow$ every poly of deg ≤ 7 will be coprime to $m(x)$, i.e. $\gcd(p(x), m(x)) = 1$ $\quad p(x)$

Run the Euclidean algorithm.

First need to find $q_1(x), r_1(x)$

$$m(x) = q_1(x)\, p(x) + r_1(x)$$

where $0 \leq \deg r_1(x) < \deg p(x) = 5$

in $GF(2^8)$ coefficients are binary

$1+1 = 0$
$1+0 = 1$
$0+1 = 1$
$0+0 = 0$
$-1 = +1$

$m(x) = x^8 + x^4 + x^3 + x + 1$

$x^3 p(x) = x^8 + x^6 + x^5 + x^3$

$x^6 + x^5 + x^4 + x + 1$

$\Rightarrow m(x) = x^3 p(x) + x^6 + x^5 + x^4 + x + 1$

$x p(x) = x^6 + x^5 + x^4 + x + 1$

$x^6 + x^4 + x^3 + x$

$x^5 + x^3 + 1$

$\Rightarrow m(x) = x^3 p(x) + x p(x) + x^5 + x^3 + 1$

$= x^3 p(x) + x p(x) + p(x) + x^2$

$$= \underbrace{(x^3 + x + 1)}_{q_1(x)}\, p(x) + \underbrace{x^2}_{r_1(x)}$$

Next we seek.

$$p(x) = q_2(x)\, r_1(x) + r_2(x)$$

$$0 \le \deg r_2(x) < \deg r_1(x) = 2$$

$$x^5 + x^3 + x^2 + 1 = \underbrace{(x^3 + x + 1)}_{q_2(x)} \cdot x^2 + \underbrace{1}_{r_2(x)}.$$

So the E.A. stops here and says $\gcd(m(x), p(x)) = 1$, as required.

For the inverse algo, if we run the extended E.A. to find the polys $a(x), b(x)$ giving Bezout's identity.

$$\boxed{a(x)\, m(x) + b(x)\, p(x) = 1}$$

and then $p^{-1}(x) = b(x)$.

Begin with the final poly division equation rewrite it as.

$$1 = p(x) + (x^3+x+1)\cdot x^2$$

and from the first poly. div. equation we have.

$$x^2 = m(x) + (x^3+x+1)\cdot p(x)$$

Substituting this into the previous gives.

$$1 = p(x) + (x^3+x+1)\left[m(x) + (x^3+x+1)\,p(x)\right]$$

$$= \underbrace{(x^3+x+1)}_{a(x)}\,m(x) + \underbrace{\left(1 + (x^3+x+1)\cdot(x^3+x+1)\right)}_{b(x)}p(x)$$

where

$$b(x) = 1 + x^6 + x^4 + x^3 + x^4 + x^2 + x + x^3 + x + 1$$

$$= x^6 + x^2$$

So $1 = (x^3+x+1)\,m(x) + (x^6+x^2)\cdot p(x)$

from this we see that

$$p^{-1}(x) = x^6 + x^2 .$$